Reading: Tools for visualizing data

In this course, you'll work with Tableau and spreadsheets. Both of these tools have advantages and disadvantages Often, data analysts will discover they need to use multiple tools, even on a single project. What you use will largely be determined by the work you're doing and your goals. This reading explores two of the tools you might use to visualize and present data: spreadsheets and Tableau.

Spreadsheets

Google Workspace and Microsoft Office Suite both offer spreadsheet applications. You've worked with Google Sheets in this course, and it's very similar in function to Microsoft Excel. If you want to compare some of the features of Sheets to Excel, check out the Microsoft video [Create a chart from start to finish](#).

Both Sheets and Excel are go-to choices for creating static charts and graphs. They offer basic data visualization capabilities that are often enough for simple visualizations. In addition, you can use them to clean, sort, and filter data. And both offer a range of chart types, graphing tools, and pivot tables for creating effective data visualizations. These charts are easy to manage; they update when the source data is updated, so they don’t require much manual intervention once implemented.

Sheets and Excel are connected to other apps in their product suites. Google Docs and Slides are very similar to Microsoft Word and Powerpoint, for example. You can incorporate data visualizations from Sheets or Excel into reports and documents in Docs and Word. Presentation programs such as Slides and Powerpoint allow you to create engaging presentations that include data visualizations so you can share insights in a presentation format. Learn more about the power of this interconnectivity among Google tools in the article Link a chart, table, or slides to Google Docs or Slides.

Tableau

Tableau is used to create powerful and interactive visualizations, making it an excellent choice for data visualizations such as live dashboards. Tableau also makes it easy to create charts, graphs, and dashboards in a drag-and-drop interface. The application supports a wide range of data sources and provides advanced analytics capabilities. These features allow for in-depth exploration of data trends and patterns.

Tableau is particularly useful for creating visualizations using huge datasets, like in this World Happiness Report by Sustainable Development Solutions which uses global reporting data on different countries' happiness ratings. Likewise, this visualization of Population and Housing State Data from 2020 United States Census Data compares population rates in the United States and available housing.

Tableau is widely known and used for its versatility and power, but it can take quite a bit of time to learn to use Tableau effectively. Soon, you’ll begin practicing with Tableau. But if you’d like to check it out now, there is a free environment you can access at Tableau Public.

Key takeaways

There are many visualization tools you will have the opportunity to use as a data professional. Different tools have different advantages and disadvantages. Although Tableau ultimately has more power than a basic spreadsheet application, it’s most often used for specific cases and to work with large datasets. Don’t underestimate how much you can do with spreadsheets or how powerful interconnectivity between apps can be!

Most of the time, especially for something like a quick report, you're more likely to reach into your toolkit for your spreadsheet app of choice. But your data career will definitely benefit from Tableau, so as you progress take advantage of opportunities to work with it. With so many different data analysis situations, familiarity with all of these tools will help you know which is the best for each situation.